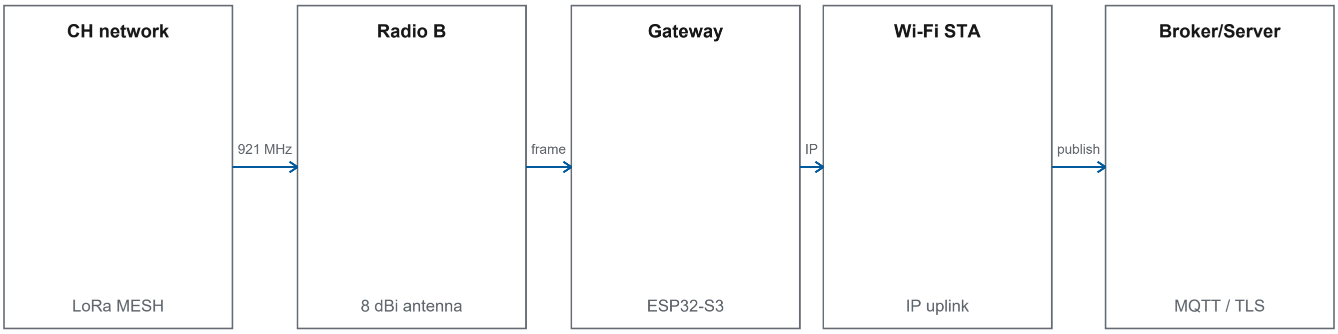
Gateway

Technical Datasheet Revision 4.0

LoRa MESH to IP-network bridge over Wi-Fi STA and MQTT, with separate TLS and non-TLS firmware.



BOARDS	ACTIVE RADIO	NETWORK	TRANSPORT
Rectangle / Circle	MESH / Radio B	Wi-Fi STA	MQTT TLS / non-TLS

Key features

- Receives frames from CH through Radio B/MESH.
- Functionally publishes uplink, status, and topology over MQTT.
- Receives commands from MQTT and forwards them as MESH downlink.
- Provides separate TLS and non-TLS firmware targets.

Technical overview

Gateway connects the LoRa MESH backbone to an MQTT broker over Wi-Fi while maintaining a downlink path to the field.

Functional profile

Ingress	LoRa MESH
IP uplink	Wi-Fi STA
Messaging	MQTT
Security option	CA-verified TLS

Operational interfaces

Uplink	MESH frame is received, processed, then published over MQTT.
Downlink	MQTT command is translated into the MESH path toward CH.
Trust boundary	TLS, when selected, terminates at the broker; it does not span LoRa.

Contents

1 Product identification

Product	Gateway
Product status	Engineering Prototype
Document revision	4.0
Technical baseline	Firmware 0.2.0 | Protocol 0.2.0
Issuer	Lab IoT ITB

1.1 Abbreviations

Abbreviation	Meaning	Abbreviation	Meaning
CA	Certificate Authority	NVS	Non-volatile configuration storage
MESH	CH-to-Gateway backbone radio domain	STA	Wi-Fi station mode
MQTT	Message Queuing Telemetry Transport	TLS	Transport Layer Security
NTP	Network Time Protocol	WAN/LAN	Deployment IP-network segment

2 Overview and hardware

2.1 Product overview

Gateway connects the LoRa MESH backbone to an MQTT broker over Wi-Fi while maintaining a downlink path to the field.

Ingress	LoRa MESH	**IP uplink**	Wi-Fi STA
Messaging	MQTT	**Security option**	CA-verified TLS

- Receives frames from CH through Radio B/MESH.
- Functionally publishes uplink, status, and topology over MQTT.
- Receives commands from MQTT and forwards them as MESH downlink.
- Provides separate TLS and non-TLS firmware targets.

2.2 Role in the system architecture

Gateway is the boundary between the field radio network and the customer IP network.

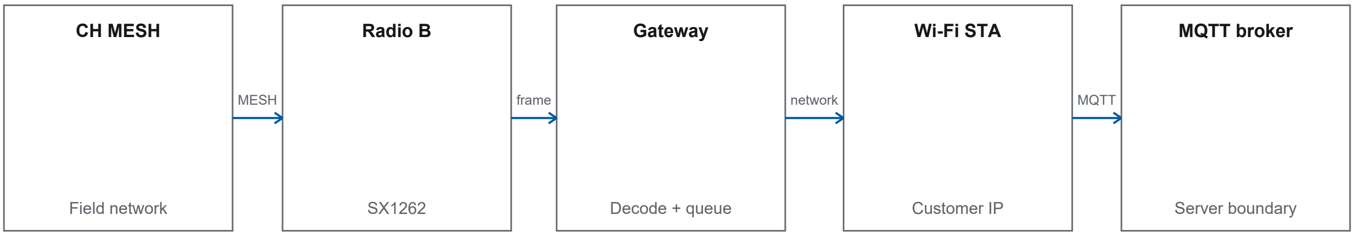


Figure 1. Gateway radio-to-IP boundary.

Uplink	MESH frame is received, processed, then published over MQTT.
Downlink	MQTT command is translated into the MESH path toward CH.
Trust boundary	TLS, when selected, terminates at the broker; it does not span LoRa.

2.3 PCB variants

Gateway uses the same PCB basis as CH but activates a different functional profile in firmware.

Product variant	Design basis	Standard selection	TLS selection
Gateway Rectangle	Rectangle hardware profile	Gateway Rectangle - standard	Gateway Rectangle - TLS
Gateway Circle	Circle hardware profile	Gateway Circle - standard	Gateway Circle - TLS

- Rectangle and Circle pin profiles differ.
- Radio A/STAR is disabled in Gateway firmware.
- Radio B/MESH is the active radio for both variants.
- Operator Hub requires board-form selection before upload.

2.4 Hardware and radio usage

Gateway uses an ESP32-S3 MCU and one active MESH radio path on a dual-radio PCB.

Function	Component / path	Firmware status
MCU	ESP32-S3-WROOM-1U-N16R8	Active
Radio A	E22-900MM22S	Disabled / unused
Radio B	E22-900MM22S	Active as LoRa MESH
Wi-Fi	ESP32-S3 station interface	Active as native IP uplink
Status LED	Board-specific mapping	Firmware-controlled

2.5 Firmware target matrix

Four targets keep board-form and MQTT-transport choices explicit.

Board	Transport	Operator Hub selection	Security profile
Rectangle	MQTT without TLS	Gateway Rectangle - standard	No
Circle	MQTT without TLS	Gateway Circle - standard	No
Rectangle	MQTT over TLS	Gateway Rectangle - TLS	Yes
Circle	MQTT over TLS	Gateway Circle - TLS	Yes

Stage	Description
Board selection	Rectangle or Circle
Transport selection	TLS or non-TLS
Firmware variant	Must match the board and transport
Device verification	Board variant, transport profile, and TLS capability are read

Figure 2. Two-dimensional Gateway target selection.

3 LoRa MESH interface

3.1 LoRa MESH interface

Gateway shares MESH defaults with CH and rejects carriers outside 920-923 MHz.

Parameter	Firmware default	Status
Carrier	921.0 MHz	Accepted 920-923 MHz inclusive
Bandwidth	125 kHz	Must match CH MESH
Spreading factor	SF9	MESH default
Coding rate	4/5	MESH default
TX setting	17 dBm	Not measured output
Antenna	8 dBi	Product configuration

> Note: The nominal 17 dBm + 8 dBi calculation yields 25 dBm before cable and connector losses.

3.2 MESH configuration and readback

Radio configuration is stored in NVS, verified, and reloaded at boot.

Stage	Description
Engineer input	MESH parameters
Validation	Carrier 920-923 MHz and other fields valid
NVS write	Configuration stored
Readback	Values are read and compared
Runtime	Valid configuration is used

Figure 3. Gateway MESH configuration cycle.

> Note: Device readback must be recorded after commissioning because NVS can override firmware defaults.

4 IP networking and MQTT

4.1 Wi-Fi network interface

The Gateway native network interface is Wi-Fi station.

Mode	Wi-Fi STA	**Uplink**	IP network
Protocol	MQTT	**Native Ethernet**	No

Provisioning	SSID and network credentials are configured on the device.
Site dependency	DHCP/static policy, VLAN, NAC, DNS, NTP, and firewall are customer-defined.
Wired option	An external Wi-Fi-to-Ethernet converter may be considered as a deployment accessory; it is not a native product interface.

4.2 MQTT functions

MQTT carries uplink/status/topology information to the broker and commands back to the Gateway.

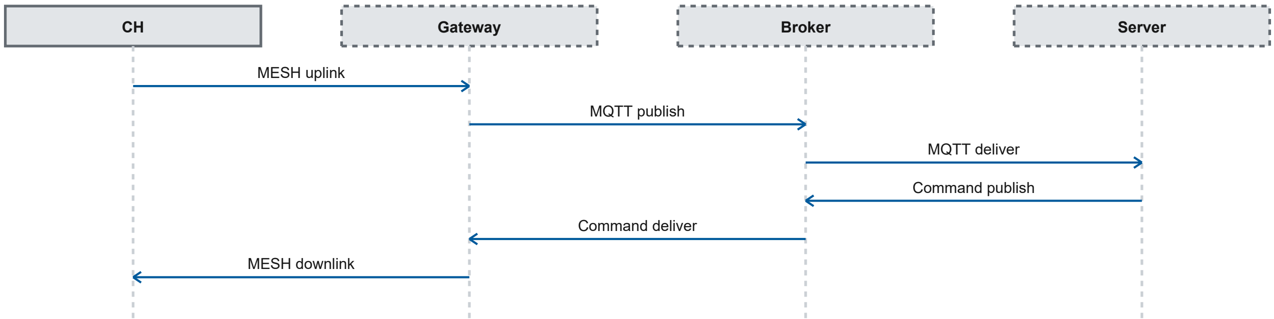


Figure 4. MQTT uplink and downlink.

> Note: Uplink publication and command reception operate after the broker session is established.

4.3 MQTT over TLS

The TLS profile uses a TLS client, Root CA, and trusted time before broker connection.

Stage	Description
TLS-capable firmware	Gateway Rectangle - TLS or Gateway Circle - TLS
Root CA	PEM must be provisioned and pass validation
Trusted time	Connection is blocked until the NTP host is valid and system time is ready
Secure client	Supports CA verification; no insecure mode is provided
Broker session	Supports MQTT over TLS after all gates are valid

Figure 5. MQTT/TLS connection gates.

- Firmware blocks TLS when Root CA or trusted time is invalid.
- TLS protects the Gateway-to-broker segment, not the entire radio path.
- Broker connection, CA validity, and trusted-time readiness are available through device readback.

4.4 Non-TLS firmware

The non-TLS target is retained as separate firmware and is not replaced by the TLS variant.

Area	Non-TLS	TLS
Board support	Rectangle + Circle	Rectangle + Circle
MQTT transport	TCP without TLS	CA-verified TLS
Root CA	Not used	Required
Trusted time	Not a TLS gate	Required for TLS
Operator selection	Explicit	Explicit

> Note: Select the TLS profile for deployments requiring encrypted broker transport and certificate verification.

4.5 MQTT queue and offline behavior

Gateway provides a bounded RAM queue for publications when MQTT is unavailable.

Capacity	8 publications	**Payload**	1-1023 bytes
Medium	RAM	**Persistence**	Volatile

Stage	Description
Publish attempt	Send directly when broker is available
Broker unavailable	Queue when space is available
Queue full	New item is rejected
Restart/power loss	Queue contents are lost

Figure 6. MQTT buffering behavior.

4.6 Command downlink path

Gateway receives commands from the broker, validates the functional target, then forwards them over MESH.

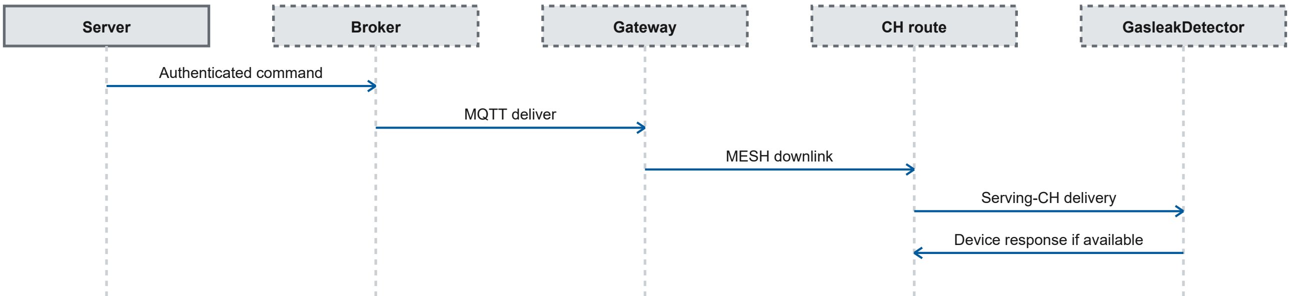


Figure 7. Command path through the Gateway.

5 Configuration and commissioning

5.1 Transactional configuration storage

Wi-Fi/MQTT configuration uses a two-slot NVS journal with readback before committing the final selector.

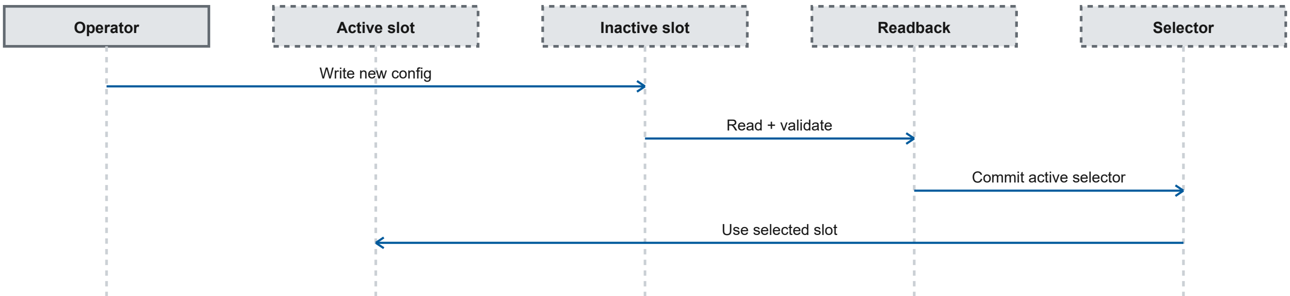


Figure 8. Two-slot configuration journal.

- Firmware provides fallback/recovery when a slot is invalid.

5.2 Provisioning through Operator Hub

Operator Hub locks board, transport, firmware variant, and TLS material before upload/configuration is accepted.

Gate	Check
Board	Rectangle or Circle must be selected.
Transport	TLS or non-TLS must be selected.
Firmware	Firmware version and variant compatibility are checked.
TLS material	Root CA PEM and NTP host are required for TLS targets.
Device verification	Board variant, transport profile, and TLS capability are read back.

> Note: TLS material is rejected on non-TLS firmware so the operator cannot mistake the active transport for a secure one.

5.3 Network deployment prerequisites

Customer deployment requires network and broker decisions outside Gateway firmware.

Area	Required deployment input
Wi-Fi	SSID, authentication, address policy, VLAN/NAC, and coverage.
Name/time	DNS and NTP reachable by the TLS target.
Broker	FQDN/IP, port, credentials, CA chain, ACL, and topic policy.
Firewall	Egress/return traffic for Wi-Fi, DNS, NTP, and MQTT/TLS.
Commissioning record	CONNACK, publish/subscribe, reconnect, logs, and certificate-validation results.

> Note: Deployment acceptance covers CONNACK, publish/subscribe, reconnect, and certificate validation.

5.4 Status and diagnostics

Readback separates firmware identity, board, transport, radio, Wi-Fi, MQTT, TLS, and queue state.

Group	Engineering readback examples
Identity	Firmware/protocol version and Gateway identity
Hardware	Board variant and active-radio role
RF	Carrier, BW, SF, CR, TX setting, preamble
Network	Wi-Fi connection state and address information
MQTT	Transport mode, broker state, reconnect, and queue depth
TLS	TLS capability and CA/time readiness on TLS targets

> Note: Device status and readback expose the MQTT/TLS connection result and active queue depth.

6 Networking, TLS, and configuration

6.1 Network configuration limits

The Gateway validates the length of each configuration field before storing and activating it.

Field	Limit	Requirement
Wi-Fi SSID	1-32 bytes	Required for Wi-Fi connectivity.
Wi-Fi password	0-64 bytes	An empty value is allowed for a network that intentionally has no password.
MQTT host	1-64 bytes	The host must be set when MQTT is enabled.
MQTT user	0-32 bytes	Used according to the broker authentication policy.
MQTT password	0-64 bytes	Used according to the broker authentication policy.
MQTT port	1-65535	Zero is rejected.
NTP host	1-64 bytes	Required for the TLS variant and must not contain whitespace.
CA certificate (PEM)	256-3900 bytes	Required for certificate verification on the TLS variant.

6.2 TLS readiness gates

The TLS variant opens the MQTT connection only after all trust and time prerequisites have been satisfied.

Check	Acceptance condition	On failure
CA certificate	A 256-3900-byte PEM value with certificate begin/end markers and no trailing data after the end marker.	MQTT connection is blocked.
NTP host	Non-empty, no more than 64 bytes, without whitespace.	Time synchronization is not started and MQTT is blocked.
Trusted time	The system epoch meets the firmware threshold: 1,577,836,800 or greater.	Certificate verification and MQTT remain blocked.
Transport	The server certificate is verified using the supplied CA.	No verification-disabled mode is used.

6.3 MQTT publication queue

When immediate publication is not possible, the Gateway can retain a small number of messages in a volatile queue.

Characteristic	Value	Behavior
Queue capacity	8 messages	Up to eight deferred publications.
Item size	1024 bytes storage	Accepted payload length is less than 1024 bytes; maximum 1023 bytes.
Retention	Volatile	Queue contents are not retained across a device restart.
Full queue	New item rejected	Existing items are not overwritten by a new item.
Queue draining	One slot per service iteration	Delivery order is not guaranteed after slots are reused.

6.4 Transactional configuration update

A network change does not immediately replace the active configuration. The Gateway completes candidate storage and verification before activation.

Stage	Action	Result
1	Validate all candidate fields.	An invalid candidate is rejected before storage.
2	Write the candidate to the inactive configuration area.	The active configuration remains available during the write.
3	Read back and compare the stored result.	Activation continues only if the result is identical.
4	Activate the candidate as the latest configuration.	The active-configuration indicator is updated last.
5	Discard the candidate if any stage fails.	The previous active configuration is retained.

6.5 Gateway product variants

Gateway functionality is available on two PCB forms. Each form supports standard and TLS transport profiles.

Device form	Standard profile	TLS profile	Active radio
Rectangle (small)	MQTT without TLS	MQTT over TLS	LoRa MESH
Circle (large)	MQTT without TLS	MQTT over TLS	LoRa MESH

> Note: PCB form and transport profile are selected during firmware preparation. The LoRa STAR radio is not used in the Gateway role.

7 References and revision history

Component references support hardware and product-interface identification.

Reference	Document
R1	Espressif - ESP32-S3-WROOM-1/1U Datasheet v1.8
R2	EBYTE - E22-900MM22S module documentation
R3	OASIS - MQTT Version 3.1.1, OASIS Standard
R4	IETF RFC 5280 - Internet X.509 Public Key Infrastructure Certificate and CRL Profile

Rev	Change
3.0	Updated technical structure and specifications.